6.1 Solution: (a) From Eq. (6.47), we have

$$\begin{aligned}\Psi(\mathbf{x},t) &= \int \frac{[f(\mathbf{x}',t')]_{\rm ret}}{|\mathbf{x}-\mathbf{x}'|}d^3x' \\ &= \int\int\int \frac{\delta(x')\delta(y')\delta(t-|\mathbf{x}-\mathbf{x}'|/c)}{|\mathbf{x}-\mathbf{x}'|}dx'dy'dz' \\ &= \int_{-\infty}^{+\infty}\frac{\delta(t-\sqrt{x^2+y^2+(z-z')^2}/c)}{\sqrt{x^2+y^2+(z-z')^2}}dz'.\end{aligned}$$

Let

$$f(z') = t - \frac{\sqrt{\rho^2+(z-z')^2}}{c},$$

with $\rho^2 = x^2+y^2$, then the roots of $f(z')=0$ are

$$z'_\pm = z \pm \sqrt{c^2t^2-\rho^2},$$

when $ct \geq \rho$. Also,

$$f'(z'_\pm) = \frac{\mp\sqrt{c^2t^2-\rho^2}}{c^2t},$$

then we can write the delta function as

$$\delta(f(z')) = \Theta(ct-\rho)\sum_\pm \frac{\delta(z'-z'_\pm)}{|f'(z'_\pm)|} = \frac{c^2t\Theta(ct-\rho)}{\sqrt{c^2t^2-\rho^2}}\sum_\pm \delta(z'-z'_\pm).$$

We can use this to calculate $\Psi(\mathbf{x},t)$ and we will have

$$\begin{aligned}\Psi(\mathbf{x},t) &= \frac{c^2t\Theta(ct-\rho)}{\sqrt{c^2t^2-\rho^2}}\sum_\pm\int_{-\infty}^{+\infty}\frac{\delta(z'-z'_\pm)}{\sqrt{\rho^2+(z-z'_\pm)}}dz' \\ &= \frac{c^2t\Theta(ct-\rho)}{\sqrt{c^2t^2-\rho^2}}\cdot\frac{2}{ct} \\ &= \frac{2c\Theta(ct-\rho)}{\sqrt{c^2t^2-\rho^2}}.\end{aligned}$$

(b) Similarly,

$$\begin{aligned}\Psi(\mathbf{x},t) &= \int \frac{[f(\mathbf{x}',t')]_{\rm ret}}{|\mathbf{x}-\mathbf{x}'|}d^3x' \\ &= \int \frac{\delta(x')\delta(t-|\mathbf{x}-\mathbf{x}'|/c)}{|\mathbf{x}-\mathbf{x}'|}d^3x' \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\frac{\delta(t-\sqrt{x^2+(y-y')^2+(z-z')^2}/c)}{\sqrt{x^2+(y-y')^2+(z-z')^2}}dy'dz'.\end{aligned}$$

Make a variable change $y' \to y'-y$ and $z' \to z'-z$, the integral becomes

$$\Psi(\mathbf{x},t) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\frac{\delta(t-\sqrt{x^2+y'^2+z'^2}/c)}{\sqrt{x^2+y'^2+z'^2}}dy'dz' = 2\pi\int_0^{+\infty}\frac{\delta(t-\sqrt{x^2+\rho'^2}/c)}{\sqrt{x^2+\rho'^2}}\rho'd\rho'.$$

Again, let

$$f(\rho') = t - \frac{\sqrt{x^2 + \rho'^2}}{c},$$

its root is

$$\rho'_0 = \sqrt{c^2t^2 - x^2},$$

when $ct \geq |x|$. Also,

$$f'(\rho'_0) = -\frac{\sqrt{c^2t^2 - x^2}}{c^2 t}.$$

Then,

$$\delta(f(\rho')) = \Theta(ct - |x|)\frac{\delta(\rho' - \rho'_0)}{|f'(\rho'_0)|} = \frac{c^2 t\Theta(ct - |x|)}{\sqrt{c^2t^2 - x^2}}\delta(\rho' - \rho'_0),$$

and

$$\begin{aligned}
\Psi(\mathbf{x}, t) &= 2\pi\frac{c^2 t\Theta(ct - |x|)}{\sqrt{c^2t^2 - x^2}}\int_0^{+\infty}\frac{\delta(\rho' - \rho'_0)}{\sqrt{x^2 + \rho'^2}}\rho' d\rho' \\
&= 2\pi\frac{c^2 t\Theta(ct - |x|)}{\sqrt{c^2t^2 - x^2}}\frac{\rho'_0}{\sqrt{x^2 + {\rho'_0}^2}} \\
&= 2\pi\frac{c^2 t\Theta(ct - |x|)}{\sqrt{c^2t^2 - x^2}}\frac{\sqrt{c^2t^2 - x^2}}{ct} \\
&= 2\pi c\Theta(ct - |x|).
\end{aligned}$$